CAmkES Notes and FAQ

Page Content

Sub-Pages

Scope

Everyting in here shall be considered as staging area for the handbook section seL4 Microkernel and the CAmkES Framework. Once stuff is understood well enough, move it over.

FAQ and Notes (Staging Area)

Adding more datatypes as attributes of components

CAmkES currently supports only (signed) ints as as data types for integers in the component configuration.

To support our use case with >4GB storage, a patch was created to add "off_t"; the branch is SEOS-1745-storage.

We will reach out to Data61 to figure out, how to proceed with this patch.

seL4NotificationQueue

descibe me and add this at seL4 Microkernel and the CAmkES Framework#Connectors

seL4GlobalAsynchCallback

descibe me and add this at seL4 Microkernel and the CAmkES Framework#Connectors

Single-Threadded components

The file sel4_global_components/components/modules/single-threaded/camkes-include/camkes-single-threaded.h defines

```
#define single_threaded_component() \
    control; \
    attribute int single_threaded = 1;
```

So a component can be declared as

```
componen foo {
    single_threaded_component()
    ...
}
```

ToDo: check that this realy does not generare separate PRC handle treands then ?

Component Header files

- see https://sel4.systems/Info/CAmkES/Tutorial.pml

Global components

- see https://github.com/seL4/global-components

Languare Specific

C

- main language
- in theory things are agnostic of a runtime library, but practically thing are married to muslibc, as this is the only lib that is actively used. Some experiments have been done with dietlibc

CakeMl

- it's supported, somebody to explore this...

Rust

- it stuck somewhere in an experimental stage, somebody to try it out

Ada

- there should be no real blocker to make this work, somebody to try it out

Architecture specific

ARM

- nothing special so far

RISC-V

- nothing special so far, works basically like on ARM
- RISC-V support was added with https://github.com/seL4/camkes-manifest/commit/f839e6fecaeacf865686d3a6fdd7391277fa7463

x86

- there are no memory mapped peripheral, but I/O ports, so drviers are slightly different